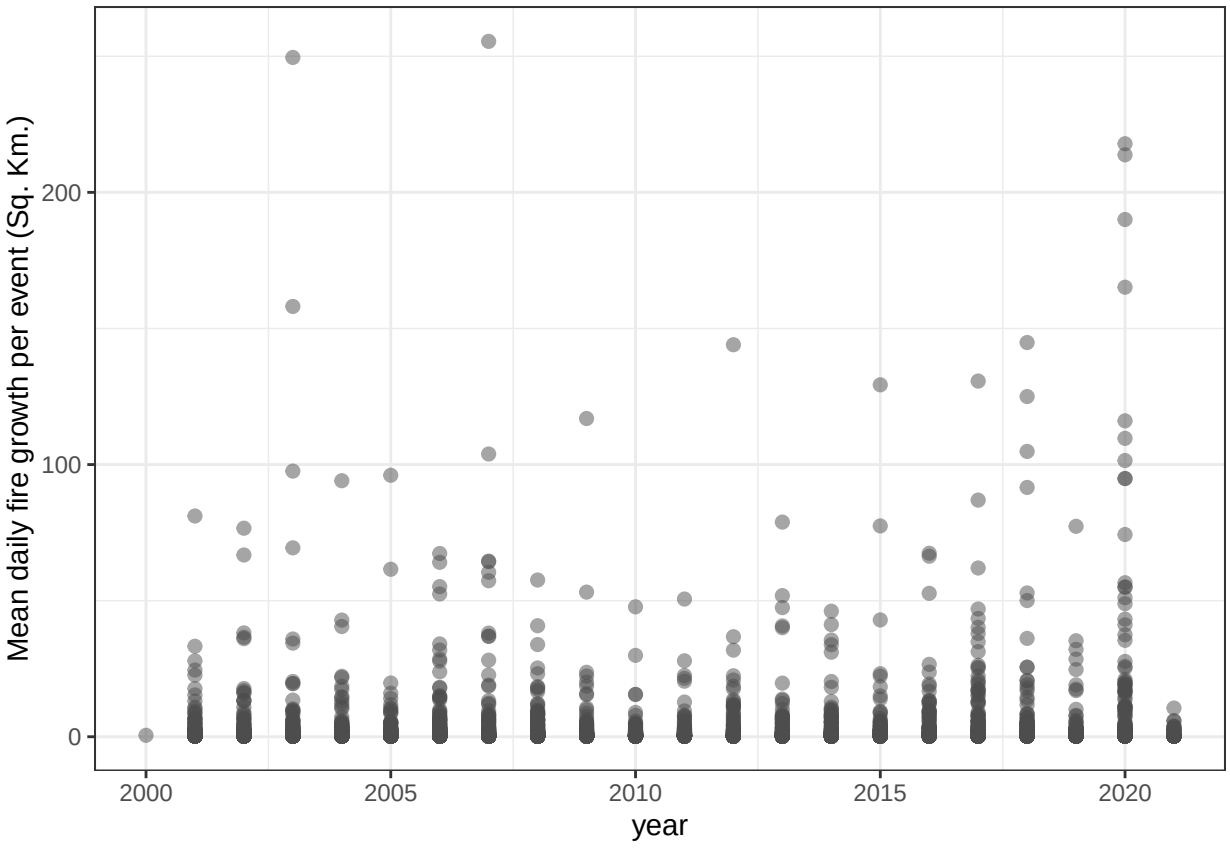
Exploratory Figure

2026-10-25

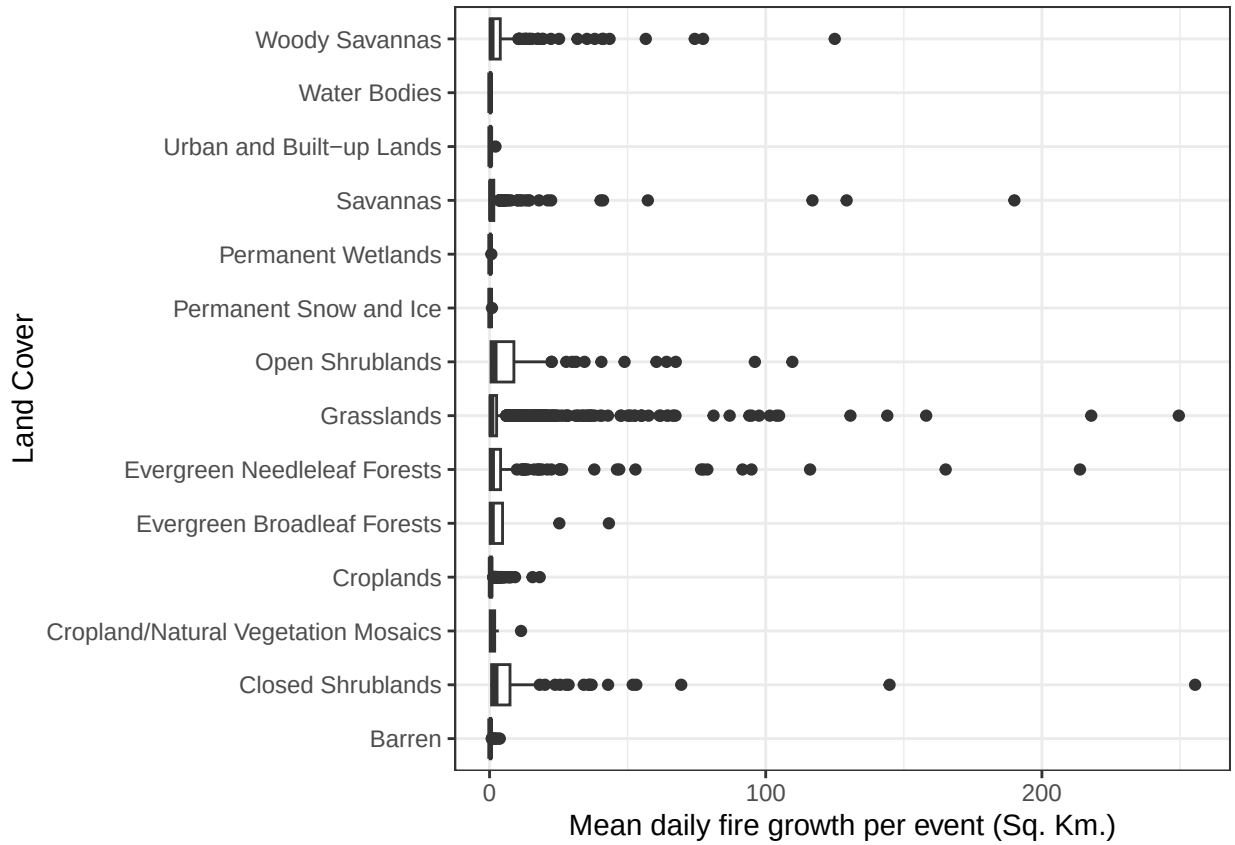
```
california_events_processed <- read_csv(here("Data/Processed/california_events_processed.csv"))
```

```
## Rows: 4329 Columns: 22
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (6): lc_name, lc_desc, lc_type, eco_mode, eco_type, eco_name
## dbl  (13): id, ig_day, ig_month, ig_year, event_dur, tot_ar_km2, fsr_km2_dy,...
## date  (3): ig_date, last_date, mx_grw_dte
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ggplot(california_events_processed, aes(x = ig_year, y = mu_grw_km2))+
       geom_point(size = 2, alpha = 0.5, color = "grey30")+
         labs(
    x = "year",
    y = "Mean daily fire growth per event (Sq. Km.)",
  ) +
  theme_bw()
```




```
ggplot(california_events_processed, aes(x = lc_name, y = mu_grw_km2))+
       geom_boxplot()+
         labs(
    x = "Land Cover",
    y = "Mean daily fire growth per event (Sq. Km.)",
  ) +
  coord_flip() +
  theme_bw()
```



Data Quality Audit The original event data source in the Scripts/ImportingScript.Rmd shown in the line stating event_data <- read_sf(here("./Data/Raw/Raw_large/fired_uscan_to2021121_events_gpkg_shp/fired_uscan_to20 was trimmed to the California boundary within ArcGIS. After importing the event data along with the California boundary found in Data/Raw/california.gpkg, the tool "Select Layer By Location" was used to select the input of events that intersected with the California boundary. Next, the fire event feature layer attribute table was opened and the selected items was reversed so that it only highlighted what was not intersecting. Finally, the selection of non-intersecting polygons of fire events were deleted from the events feature layer and exported to Data/Raw/california_events.shp.

```
# Data Quality Audit

event_data <- read_sf(here("./Data/Raw/Raw_large/fired_uscan_to2021121_events_gpkg_shp/fired_uscan_to20
dim(event_data)
```

```
## [1] 130761     30
```

```
california_events <- read_sf(here("./Data/Raw/California_events.shp"))
dim(california_events)
```

```
## [1] 4329   32
```

Trimming the data set to California narrowed the amount of observations from 130,761 to 4,329 and added the shape length and shape area columns for the polygons. The temporal coverage for the California events is from 12/11/2000 to 06/01/2021. For the primary cleaning of this data, the columns containing datetime information were transformed from character to date and the land cover/ ecosystem names were transformed

from character to factor. Then the columns that specified the event growth in pixels were excluded from the processed dataframe, since their equivalent in square kilometers was given and ultimately easier to interpret. Also, the unnecessary shape length and shape area columns from ArcGIS along with the geospatial columns (x,y) were removed. There are three NAs in the dataframe within two different event rows. They are missing information for either eco_name or both eco_mode and eco_name. We have decided to keep these because they don't affect analysis for land cover and can be dropped later if needed. There are no duplicate rows.

```
# number of NA values in dataframe = 3
sum(is.na(california_events_processed))
```

```
## [1] 3
```

```
which(is.na(california_events_processed))
```

```
## [1] 77978 78705 86636
```

```
california_events_processed[!complete.cases(california_events_processed),]
```

```
## # A tibble: 2 x 22
##      id ig_date    ig_day ig_month ig_year last_date  event_dur tot_ar_km2
##   <dbl> <date>      <dbl>    <dbl>   <dbl> <date>         <dbl>      <dbl>
## 1  2367 2001-07-07    188        7    2001 2001-07-07         1      0.215
## 2 26000 2004-07-05    187        7    2004 2004-07-05         1      0.429
## # i 14 more variables: fsr_km2_dy <dbl>, mx_grw_km2 <dbl>, mn_grw_km2 <dbl>,
## #   mu_grw_km2 <dbl>, mx_grw_dte <date>, lc_code <dbl>, lc_mode <dbl>,
## #   lc_name <chr>, lc_desc <chr>, lc_type <chr>, eco_mode <chr>,
## #   eco_type <chr>, eco_name <chr>, tot_perim <dbl>
```

```
# number of duplicated rows = 0
sum(duplicated(california_events_processed))
```

```
## [1] 0
```